

# Attention Research

Results across two projects

## PROJECT 1

### CRPA: is behavioural contribution measurable at all?

*Audited a partitioned-attention repository, then tested whether its own central claim survives a resolvability check.*

#### 1.1 The claim that was withdrawn

The branch reported that structural overlap does not predict behavioural contribution. Check 0 measures each edge's effect twice on disjoint halves of the evaluation split and correlates the two estimates. **Pooled verdict: UNRESOLVABLE.**

| seed | r_delta | r_stat | ceiling | observed rho | delta size | converges |
|------|---------|--------|---------|--------------|------------|-----------|
| 42   | +0.088  | +0.119 | 0.102   | +0.018       | 6.0 ULP    | no        |
| 1337 | -0.026  | +0.255 | 0.000   | -0.017       | 5.0 ULP    | no        |
| 2024 | +0.012  | +0.250 | 0.054   | +0.013       | 4.0 ULP    | no        |

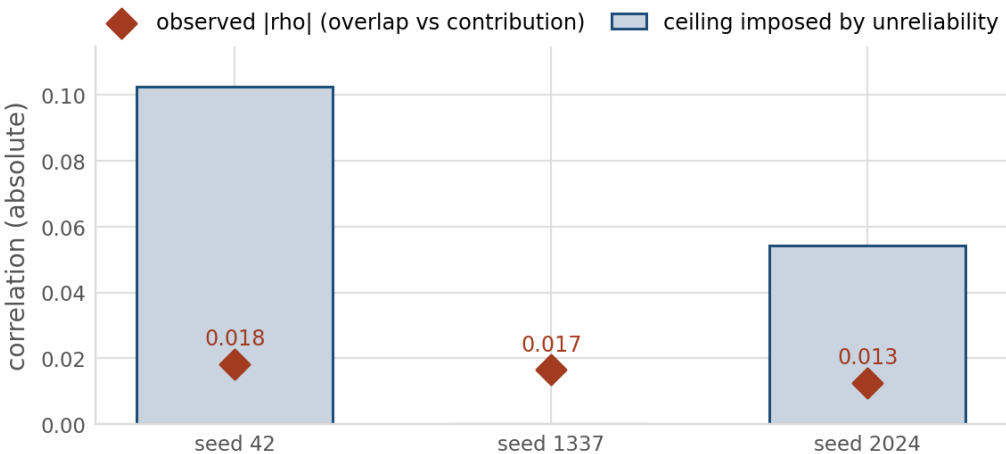


Figure 1. Observed correlation against the ceiling that measurement unreliability imposes on it,  $\sqrt{r_{\text{delta}} \times r_{\text{stat}}}$ . Every observed value sits far inside its own ceiling, so a real decoupling and measured noise cannot be told apart. Seed 1337 has no bar: its split-half reliability is negative, so the ceiling degenerates to zero and that seed carries no information at all.  $n = 3$  seeds, 1,044 to 1,152 scored edges each.

## 1.2 The floor holds across three orders of magnitude

The replacement claim is that single-edge contribution is not resolvable, not that overlap fails to predict it.

| scale          | precision | single-edge delta                           |
|----------------|-----------|---------------------------------------------|
| 12.4M (Tier 1) | float32   | 4 to 6 ULP, split-half reliability 0.088    |
| 138M (Tier 2)  | float32   | single-edge p90 0 ULP; group of 24 is 1 ULP |
| 6.9B (Tier 3)  | bfloat16  | exactly 0.000e+00 across 192 edges          |

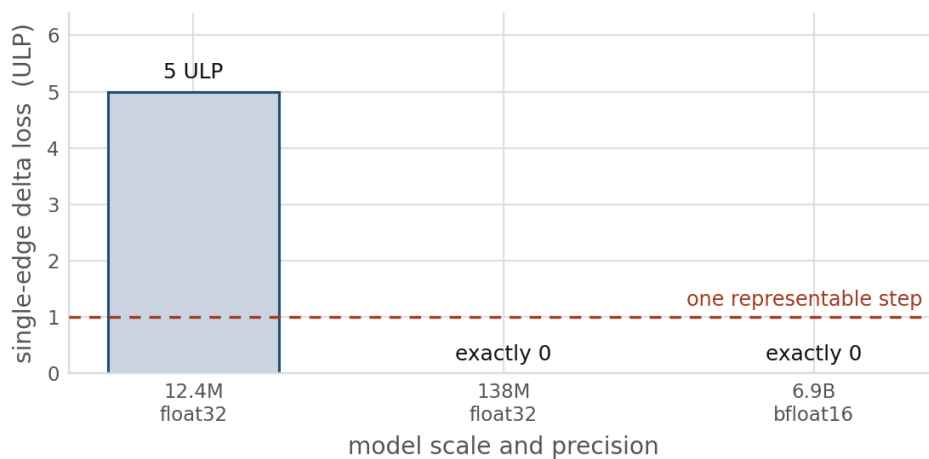


Figure 2. Typical single-edge intervention effect, expressed in units of the least representable step at the working precision. At 138M the single-edge p90 is zero; removing all 24 candidates together moves the loss by one ULP. At 6.9B the single-edge effect is exactly zero: the edit shifts logits by 2.58, 3.05 and 0.127 at layers 0, 16 and 31, and the loss is bit-identical to baseline for all 192 edges. A criterion thresholding this quantity is unfalsifiable at every scale tested.

## 1.3 Reproduction of the published table

Re-run from the original commit. Four of five rows come back. The one that does not is the one the paper's claim rests on.

| Variant                    | Published | Reproduced | Verdict            |
|----------------------------|-----------|------------|--------------------|
| Dense Transformer          | 50.9%     | 55.6%      | reproduces         |
| Sliding Window             | 51.9%     | 52.5%      | reproduces         |
| CRPA no regularisation     | 8.4%      | 10.9%      | reproduces         |
| CRPA naive regularisation  | 5.3%      | 5.3%       | reproduces exactly |
| CRPA causal regularisation | 32.8%     | 7.5%       | does not reproduce |

## 1.4 Three seeds against a measured chance floor

The floor is measured by simulating the task generator, not asserted from vocabulary size: 52.78% (sd 0.08, n = 25,000 x 5). **No variant clears it, dense included.** The retrieval task cannot discriminate between these variants.

| Variant           | Retrieval | t vs floor | Verdict           | Overlap |
|-------------------|-----------|------------|-------------------|---------|
| dense             | 53.6%     | +0.8       | indistinguishable | 0.403   |
| sliding window    | 47.2%     | -1.4       | indistinguishable | 0.322   |
| crpa_noreg        | 4.4%      | -66        | below             | 0.244   |
| crpa_naive        | 4.6%      | -100       | below             | 0.264   |
| crpa_contribution | 4.3%      | -50        | below             | 0.222   |

## 1.5 Long context: measured cost, and a proven wall

A fused dense kernel is faster at every length measured, at equal memory, with the gap narrowing as context grows.

| Context | Dense   | Sliding  | CRPA     | CRPA vs dense  | Peak memory          |
|---------|---------|----------|----------|----------------|----------------------|
| 4,096   | 13.2 ms | 19.5 ms  | 31.5 ms  | 2.40x slower   | 737 MB               |
| 8,192   | 30.0 ms | 62.4 ms  | 60.1 ms  | 2.00x slower   | 1,181 MB             |
| 16,384  | 72.0 ms | 218.1 ms | 119.1 ms | 1.65x slower   | 2,067 MB             |
| 32,768  | –       | –        | –        | <i>not run</i> | <i>out of memory</i> |
| 65,536  | –       | –        | –        | <i>not run</i> | <i>out of memory</i> |

32k and 64k were retried five times, including on an idle A100 80GB at the correct profile with batch size 1 and adaptive chunking. All exceed memory. The investigation established something more useful than the failure: **a forward pass at 32k costs 1.90 GB peak**, so the model scales and the cost sits in the diagnostic, not in inference.

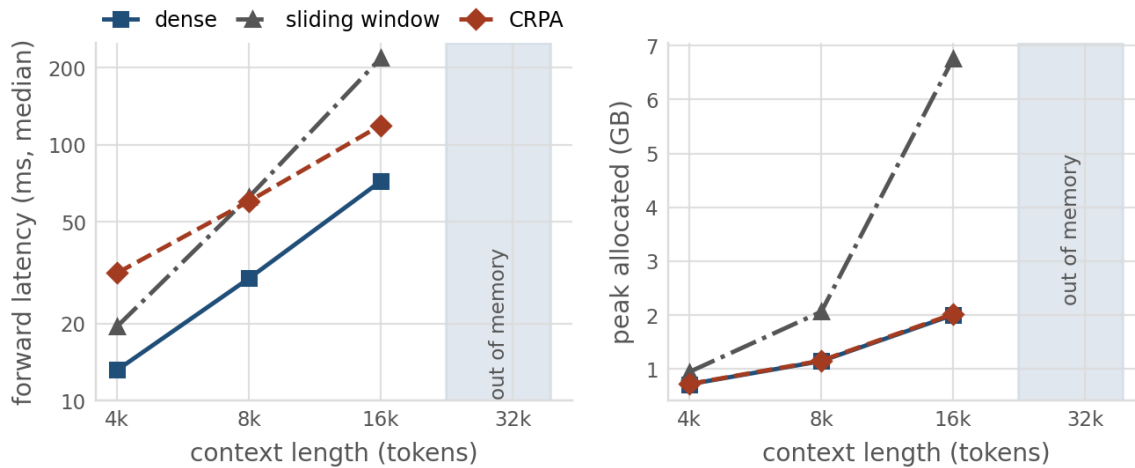


Figure 3. Measured forward latency and peak allocated memory at the three context lengths that fit. A fused dense kernel is faster than the sparse gather path at every length, and the gap narrows from 2.40x to 1.65x as context grows, so the sparse path is closing but has not overtaken it by 16k. The shaded region marks lengths that exhausted memory on every attempt and are reported as not run rather than extrapolated. Batch size 1, bfloat16, 138M parameters, median of 10 iterations after 5 warmups.

## 1.6 Defects found and repaired

Each of these changed a reported number.

| Defect                                                        | Consequence                                                      |
|---------------------------------------------------------------|------------------------------------------------------------------|
| Central claim unsupported by its own measurement              | A decoupling claim the data cannot support. Withdrawn            |
| Chance floor asserted, not measured                           | Real floor 52.78%, not 5.0%. Every above-chance verdict inverted |
| Intervention masked a query-row pair, not a query-to-key edge | The measured object was not the object being reasoned about      |
| Roughly half of all interventions were causally inert         | Delta zero by construction, so the pair was filed redundant      |
| Short runs recordable as completed measurements               | Nine 3-iteration runs entered an aggregate and moved it by 26x   |
| Auxiliary routing entropy pinned at $\ln(4)$                  | A constant reported as a finding                                 |

## xsa-controls: sanity checks for attention surgery

Three checks shipped as importable code: is the measurement resolvable, does the statistic beat an anisotropy null, and does a matched arbitrary intervention recover the gain?

### 2.1 Check 1 across a nine-model scale ladder

5,408 head-level rows, 32 real documents per model, eager attention, null partner drawn within the sequence from positions the query could causally attend.

| model              | params      | cos_self      | cos_null      | excess        | % self-specific |
|--------------------|-------------|---------------|---------------|---------------|-----------------|
| gpt2               | 124M        | 0.4828        | 0.2987        | 0.1840        | 38.1            |
| pythia-160m        | 160M        | 0.4180        | 0.2637        | 0.1544        | 36.9            |
| gpt2-medium        | 355M        | 0.4252        | 0.2579        | 0.1674        | 39.4            |
| pythia-410m        | 410M        | 0.4022        | 0.1937        | 0.2086        | 51.9            |
| gpt2-large         | 774M        | 0.4213        | 0.2117        | 0.2096        | 49.7            |
| pythia-1.4b        | 1.4B        | 0.3862        | 0.1900        | 0.1963        | 50.8            |
| gpt2-xl            | 1.5B        | 0.3861        | 0.2069        | 0.1792        | 46.4            |
| pythia-2.8b        | 2.8B        | 0.3565        | 0.1859        | 0.1705        | 47.8            |
| <b>pythia-6.9b</b> | <b>6.9B</b> | <b>0.3404</b> | <b>0.1979</b> | <b>0.1425</b> | <b>41.9</b>     |

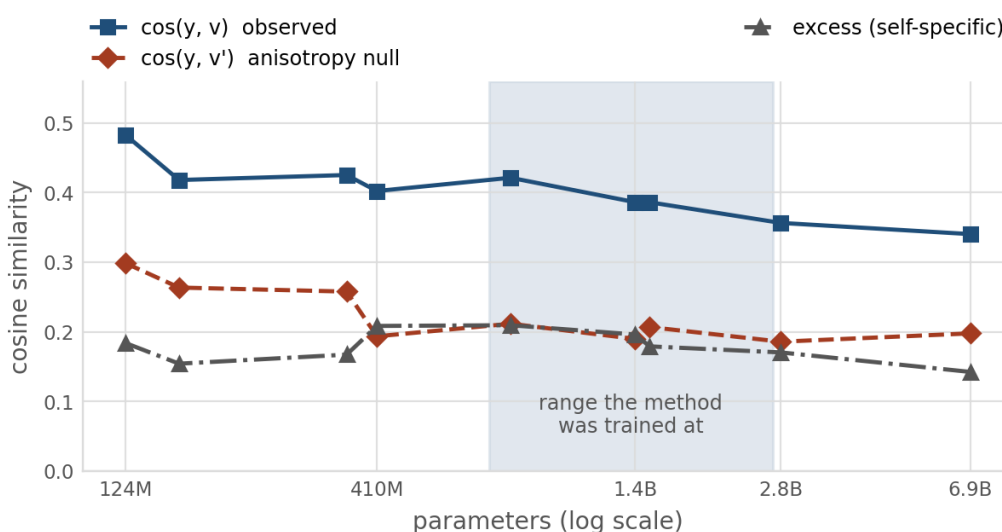


Figure 4. The anisotropy null weakens with scale but does not vanish.  $\text{cos\_null}$  falls from 0.2637 at 160M to 0.1979 at 6.9B, yet 58% of the raw statistic is still explained by the null at 6.9B, and about half across the shaded range where the method was actually trained. This answers the strongest objection to the framing, that large models are isotropic, with measurement rather than extrapolation.  $n = 9$  models, 5,408 heads.

## 2.2 The null depends on where it is measured

The same model and corpus, measured at five context lengths.

| Context length             | 64           | 128          | 256          | 512          | 1024         |
|----------------------------|--------------|--------------|--------------|--------------|--------------|
| $\cos(y, v)$ observed      | 0.4900       | 0.4846       | 0.4812       | 0.4772       | 0.4743       |
| $\cos(y, v')$ null         | 0.3747       | 0.3361       | 0.3096       | 0.2904       | 0.2788       |
| <b>self-specific share</b> | <b>23.5%</b> | <b>30.7%</b> | <b>35.7%</b> | <b>39.2%</b> | <b>41.2%</b> |

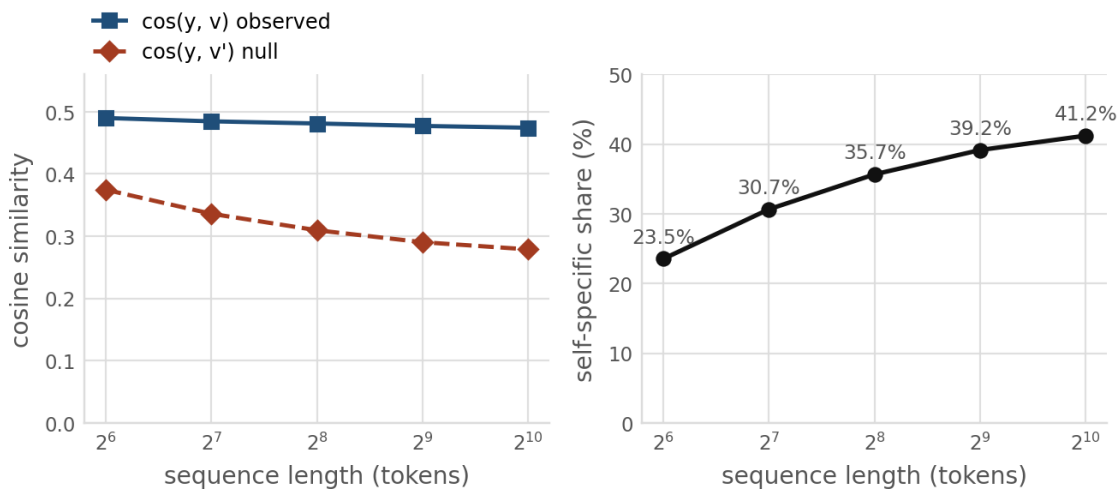


Figure 5. The observed statistic is nearly flat across a sixteenfold change in context length while the null moves by a third, so the self-specific share swings from 23.5% to 41.2%. A claim of the form “only N% of this statistic is self-specific” is therefore a property of the measurement context as much as of the model, and Check 1 must always be reported with the length it was measured at.  $n = 24$  documents per length, GPT-2.

## 2.3 Grouped-query attention, which nobody had checked

Under GQA several query heads share one KV head, so the token's own value vector is shared across a group. The source method's own table reports no KV-head count.

| model        | query / KV heads | within-group excess | across-group excess |
|--------------|------------------|---------------------|---------------------|
| Qwen2.5-0.5B | 14 / 2           | <b>+0.2415</b>      | <b>-0.1876</b>      |
| Qwen2.5-1.5B | 12 / 2           | <b>+0.2731</b>      | <b>-0.1922</b>      |

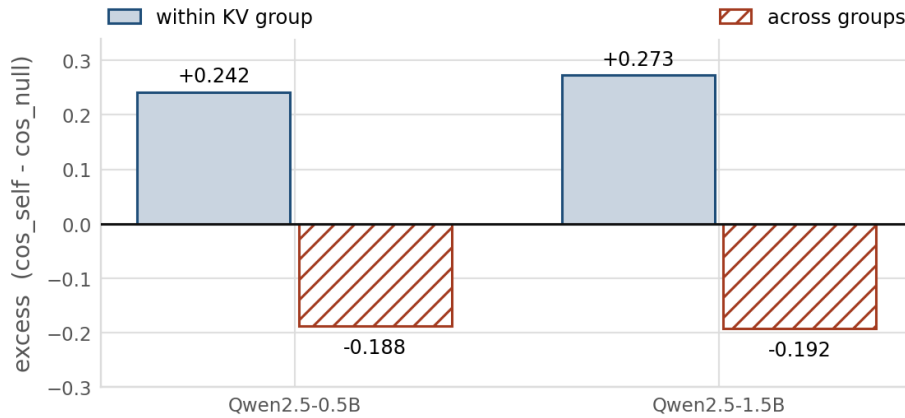


Figure 6. Borrowing a neighbouring KV group's value at the same position does not merely lose the effect, it reverses it. The self-value direction is specific to the head's own group. GQA models also show a higher self-specific share, 56 to 59%, than any multi-head model on the ladder, 37 to 52%, so grouped-query attention is not a neutral change of variable for this family of methods.  $n = 2$  models, 672 heads.

## 2.4 The matched intervention, at full seed count

24 cells, 3 arms x 8 seeds, identical initialisation and data order per seed. Measured step-0 deviation across all five arms is 0.000e+00, so the arms start from a common point.

| arm              | mean delta vs baseline | 95% CI                 | t     | p            | n |
|------------------|------------------------|------------------------|-------|--------------|---|
| random (primary) | <b>+0.001190</b>       | [+0.000351, +0.002040] | +2.48 | <b>0.042</b> | 8 |
| xsa              | +0.001515              | [-0.001223, +0.004807] | +0.92 | 0.387        | 8 |

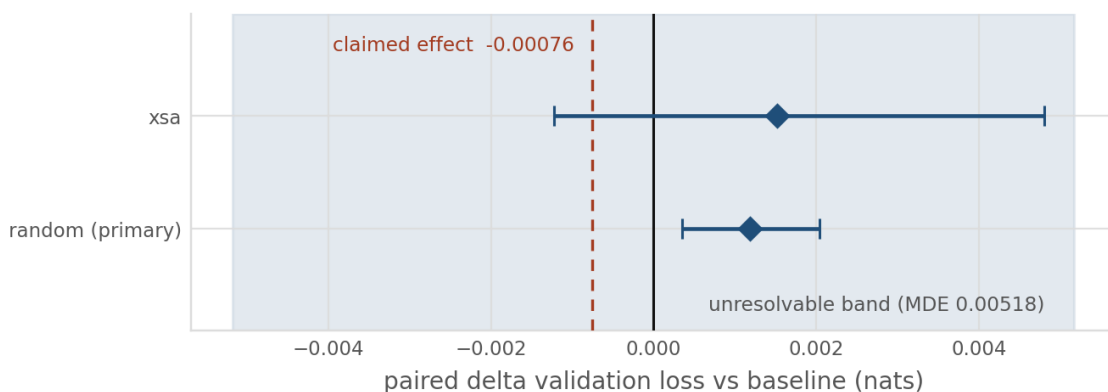


Figure 7. Both arms fall inside the shaded band of effects the design cannot resolve. The minimum detectable effect is 0.00518 nats against the 0.00076 the method's own independent replication reports, so the study is underpowered about sevenfold and cannot settle the question in either direction. The two arms are also indistinguishable from each other, which is what Check 2 asks. Reported as a power failure, not as a null result.  $n = 8$  seeds per arm.

## 2.5 The checklist discriminates between methods

Check 1 applied to two further methods on frozen GPT-2, each with a null matched to the structure the statistic inherits for free.

| method                       | statistic | null   | excess | % self-specific | survives |
|------------------------------|-----------|--------|--------|-----------------|----------|
| attention sinks              | 0.4028    | 0.0034 | 0.3994 | 99.2            | yes      |
| massive activations          | 12.8719   | 3.6452 | 9.2266 | 71.7            | yes      |
| self-value (this work, 6.9B) | 0.3404    | 0.1979 | 0.1425 | 41.9            | no       |

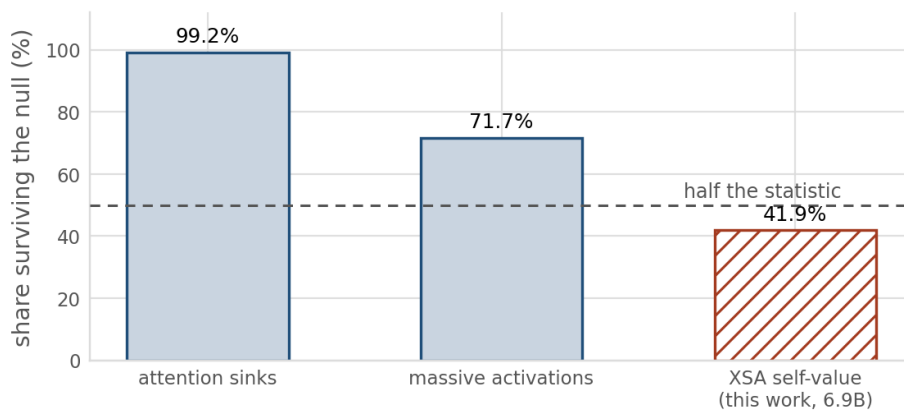


Figure 8. Attention sinks retain 99.2% of their statistic after a null matched for recency, and massive activations 71.7% against the maximum expected from a matched-variance Gaussian. The self-value statistic retains 41.9%. The checklist separates methods rather than debunking all of them, which is a stronger position than a blanket null.

## 2.6 A reproduction left failing on purpose

Published GPT-2 reference: `cos_self` 0.5406, `cos_null` 0.3798, `excess` 0.1608. Thirteen measurement conventions were tested and the full grid is reported unselected.

| convention varied                   | range of <code>cos_self</code> observed | reproduces reference |
|-------------------------------------|-----------------------------------------|----------------------|
| sequence length, 64 to 1024         | 0.4827 to 0.4955                        | no                   |
| position 0 included                 | 0.4847                                  | no                   |
| null partner definition, 3 variants | 0.4832                                  | no                   |
| head pooling                        | 0.4832                                  | no                   |
| layer subsets, 4 variants           | 0.4405 to 0.6494                        | no                   |

**No tested convention reproduces the reference.** Layer subsets move the statistic most, so a subset is the likeliest remaining explanation, but no tested subset lands on the reference triple. Reported as unexplained rather than resolved by further search: selecting the setting that hits a target and presenting it as the method is the practice this work argues against.



## 2.7 Defects found only by running it on GPU

These defects were not visible from CPU testing.

| Defect                                        | Consequence                                                         |
|-----------------------------------------------|---------------------------------------------------------------------|
| diagmask crashed on every GPU run             | bf16 autocast dtype mismatch. One of five arms could never have run |
| Calibration output was never consumed         | The factorial used a default budget, so the gate changed nothing    |
| Budget clamped up to an unaffordable floor    | Returned a plan that could not be paid for and called it fine       |
| The spend threshold was never enforced        | Defined in config, referenced nowhere                               |
| Two runs mixed token budgets in one directory | Caught by a guard ported from the sibling project; data discarded   |

## Status, and what remains open

Both deliverables carry green CI. No compute is running.

|                        | CRPA                                  | xsa-controls                                       |
|------------------------|---------------------------------------|----------------------------------------------------|
| Tests                  | 266 passing                           | 202 passing                                        |
| Coverage               | 83%                                   | 90%                                                |
| Figures from real data | 7 of 7                                | 5 of 5                                             |
| Compute used           | A6000, RTX 6000 Ada, A100 80GB        | RTX 6000 Ada, A100 80GB                            |
| Open item              | long context at 32k and 64k           | GPT-2 reference reproduction                       |
| Why it is open         | proven memory wall over five attempts | thirteen conventions tested; no convention matches |